

Electricity price dynamics in Sweden

Regression analysis of factors influencing electricity pricing

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Abstract

This thesis examines the relationship between weather variables (wind speeds and temperature), fossil fuel prices (natural gas and coal), and electricity prices in Sweden's electricity market from 2013 to 2023. This research employs multiple linear regression to quantitatively analyze how weather variables and fossil fuel prices influence electricity prices across Sweden's four electricity price areas (SE1, SE2, SE3, and SE4). The study utilized electricity price data from Nord Pool, meteorological data from the Swedish Meteorological and Hydrological Institute, and energy commodity price data from the International Monetary Fund.

The findings reveal that wind speeds have a negative correlation with electricity prices in each electricity price area analyzed. Similarly, temperature also exhibits a negative correlation with electricity prices. In contrast, natural gas and coal prices show a positive correlation with electricity prices. The explanatory power of natural gas and coal prices is higher compared to weather variables; however, this influence diminishes in the northern electricity price areas.

Addressing gaps in existing research, this thesis enhances the understanding of the energy market amid the transition towards more sustainable energy systems. The findings have implications for policy formulation and market strategy, particularly in enhancing energy security and sustainability in Sweden. Additionally, this study explores the merit order effect, emphasizing how renewable energy integration influences electricity market volatility and pricing. By providing a comprehensive analysis of these factors, this research offers valuable insights for stakeholders aiming to navigate and optimize the evolving landscape of Sweden's electricity market.

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1. Introduction

1.1 Background

Optimizing cost efficiency in electricity generation and achieving a competitive levelized cost of electricity are crucial elements of the energy markets. The levelized cost of electricity represents the average lifetime cost of generating electricity from a power plant, accounting for all expenses and future cash flows (Jacobsson, Lidström, & Öhlén, 2016). This standardized metric enables the comparison of different electricity generation sources.

Efficient and functioning electricity markets play a critical role in maintaining a consistent and sustainable energy supply, meeting the energy needs of societies, and promoting economic development and environmental sustainability (World Bank Group, 2020). This is highlighted by global initiatives such as the Paris Agreement, ratified by 196 countries, which aims to achieve net-zero emissions by 2050 (United Nations Framework Convention on Climate Change, 2016). The commitment to sustainability has propelled the growth of sustainable energy sources on a global scale. Projections indicate that by 2050, sustainable energy is expected to account for 85% of global electricity production, a significant increase from the current 30% share (International Energy Agency, 2021).

The energy crisis that unfolded in Europe in 2022 has led to high and volatile energy prices, negatively impacting households and businesses in many cases (IEA, 2023). This highlights the crucial factors of achieving affordable energy prices and securing the supply for efficient and functioning energy markets. High fossil fuel prices seen in the energy crisis have also fueled inflation pressures, combining with a high-interest-rate environment, thereby exacerbating economic challenges and increasing the risk of recession (Konjunkturinstitutet, 2022). A report commissioned by Sveriges Riksbank has concluded that despite Sweden's substantial reliance on non-fossil fuel electricity generation (over 95%), significant spikes in fossil fuel prices, particularly natural gas prices, have led to increased electricity prices in the country. These spikes have also created price disparities among different electricity price areas within Sweden (Sveriges Riksbank, 2023).

Renewable energy sources such as wind and solar power are particularly susceptible to weather-related risks due to their intermittent nature and the current lack of large-scale energy storage solutions (Jacobsson, Lidström, & Öhlén, 2016). The financial returns and sustainable benefits from these renewable sources are directly influenced by weather conditions (Creti & Fontini, 2019). As a result, financial investors in renewable energy face exposure to weather-related risks.

The market for trading weather-related financial risk has existed since the 1990s, when the first standardized weather derivatives contract was traded on the Chicago Mercantile Exchange (CME Group, 2024). Initially, weather derivatives focused on hedging temperature-related financial risks, such as those faced by utility companies during winters with extreme temperatures (Jewson & Brix, 2005). To ensure the growth of sustainable energy systems and attract investments, tools like weather derivatives for managing weather-related risks are essential. These tools are vital for expanding intermittent renewable energy in the absence of large-scale energy storage systems, by

limiting financial loss and ensuring the sustainable development of society and the energy system.

1.2 Problem discussion

Numerous studies have empirically confirmed the presence of the merit order effect in electricity markets worldwide, particularly with increased production from renewable energy sources like wind power (Clò, Cataldi, & Zoppoli, 2015; Cludius, Hermann, Matthes, & Graichen, 2014; Macedo, Marques, & Damette, 2021). The merit order effect is a concept in electricity markets used to determine the order in which different electricity generation sources are utilized to meet the power demand, based on their marginal costs of production (Jacobsson, Lidström, & Öhlén, 2016).

Macedo et al. (2021) specifically investigated the impact of rising wind power generation on Sweden's day-ahead electricity market, confirming that wind power helps reduce the levelized cost of electricity in Sweden. The correlation between wind power expansion and the comparatively lower levelized cost of electricity, driven by wind power's low marginal cost for power production compared to other energy sources, contributes positively to affordability. Therefore, further research into the dynamics of electricity prices amid wind power growth is of interest.

A study by Sandberg (2024) focused on analyzing wind speeds and temperature conditions in the Swedish electricity market between 2022 and 2023, specifically examining how measured changes in wind conditions and temperature impact electricity prices. Historically, temperature fluctuations have significantly influenced electricity prices in Sweden, particularly during cold weather months. However, with the rise of wind power, there has been a notable shift in this dynamic. The study found that a change in wind speed by 1 m/s affects electricity prices more significantly than a change in temperature by 1 degree Celsius (Sandberg, 2024). This shift underscores the changing landscape of energy production in Sweden and highlights the importance of understanding how different factors, such as wind conditions, influence electricity prices.

To examine the changes in the electricity market resulting from the expansion of sustainable energy in Sweden, historical hourly electricity price data from the Nord Pool electricity exchange will be utilized. The availability of data from 2013 enables empirical analysis with electricity prices as the dependent variable. This thesis will specifically focus on two key weather variables: wind speeds and temperature. Additionally, to contextualize the influence of weather conditions on electricity prices, variables for natural gas and coal prices will be incorporated. The study also includes control variables for the season and the economic state (GDP gap). Furthermore, the study aims to investigate all Swedish electricity markets—SE1 (Luleå), SE2 (Sundsvall), SE3 (Stockholm), and SE4 (Malmö)—to identify where the respective variables have the most significant impact. Figure 1 shows a map of Sweden displaying the different electricity price regions examined in the thesis.



Figure 1. Electricity price areas in Sweden, categorized from north to south: SE1 Luleå, SE2 Sundsvall, SE3 Stockholm, and SE4 Malmö (Fortum, 2024).

Our research question is as follows:

- (1) What is the relationship between weather variables (wind speeds & temperature) and electricity prices in the Swedish electricity market in the electricity price area SE1, SE2, SE3 and SE4?
- (2) What is the relationship between fossil fuel prices (natural gas and coal) and electricity prices in the Swedish electricity market in the electricity price area SE1, SE2, SE3 and SE4?

1.3 Aim and knowledge contribution

The study aims to empirically analyze the relationship between weather variables (including wind speeds and temperature) and fossil fuel prices (natural gas and coal) with electricity prices in Sweden from 2013 to 2023. The purpose of the study is to understand the factors influencing electricity prices through regression analysis. The study will analyze the relationships between electricity prices, weather conditions, and fossil fuel prices. Additionally, it will incorporate control variables for seasonality and the state of the economy (GDP gap). The knowledge contribution of this research is to illuminate how these factors have impacted the Swedish electricity market and how these influences vary across different electricity price areas within the country.

1.4 Limitations

This thesis only includes electricity spot price data from the Swedish electricity market, covering the price regions: SE1 Luleå, SE2 Sundsvall, SE3 Stockholm, and SE4 Malmö. Additionally, the thesis only includes weather data obtained from the weather stations: Luleå-Kalix Airport, Sundsvall-Timrå Airport, Stockholm-Bromma Airport, and Malmö Airport. A limitation of the study stems from the constrained utilization of weather data from available weather stations. Due to the abundance of data available from numerous weather stations in each price area, it is impractical to include data from all stations with hourly measurements, resulting in a considerable large sample. This clearly limits the understanding of the country's actual weather dynamics.

2. Literature review and theoretical framework

2.1 Introduction

Sweden's electricity markets have undergone notable transformation over the years. Historically dominated by hydropower and nuclear power, which accounted for 96% of production in 1990, their combined share has gradually decreased to 70% by 2023 (Energimyndigheten, 2022). This decline matches the significant growth of renewable energy sources in Sweden during the 21st century. Wind power, now a prominent contributor, accounted for 21% of total electricity production in 2023, rising from nearly 500 GWh in the 2000s to almost 35,000 GWh by 2023 (Energimyndigheten, 2024), see figure 2.

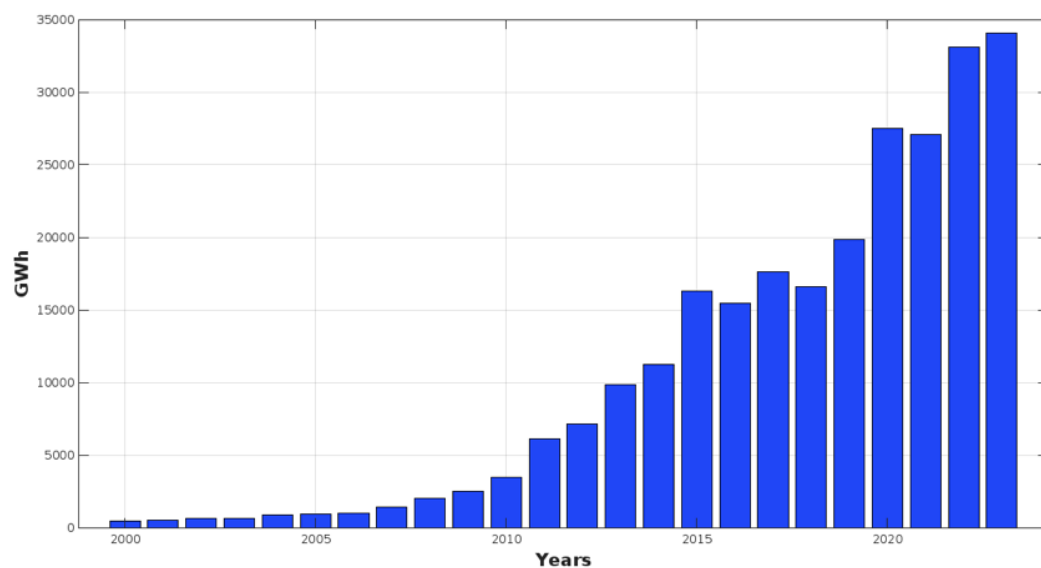


Figure 2. Annual electricity production from wind power in Sweden measured in gigawatt-hours between 2000 and 2023 (Energimyndigheten, 2024).

Renewable energy is anticipated to play a key role in the transition toward less carbon-intensive and sustainable energy systems in Sweden, as outlined by the Swedish government (Regeringen, 2019). With supportive government policies, the growth has been significant. This growth has introduced complexities to the electricity system by introducing supply variability, leading to fluctuations and spikes in electricity prices (Hagfors et al., 2016). Hagfors et al. also suggest that probability models for electricity capacity are well-suited to aid in risk management for electricity markets.

2.2 Literature survey

Previous studies on the weather's impact on energy markets mainly focused on the impact of extreme weather events on energy assets, such as hurricanes and floods. For instance, Mu (2007) found that weather shocks affected both the average and volatility of US natural gas futures prices. However, post-2010, attention shifted to weather's influence on renewable energy. Gielen et al. (2019) emphasized the economic benefits of renewable energy and its role in transitioning to sustainable energy systems by 2050, highlighting energy efficiency and renewable energy technologies as core elements of this transition. Our literature review will focus on the development of sustainable energy, particularly emphasizing research on the merit order effect and challenges associated with integrating intermittent renewable energy, cross-border power connections, and future electricity market design.

2.2.1 The Merit Order Effect

Clò et al. (2015) conducted a study to assess the merit order effect in the Italian day-ahead wholesale power market. Using hourly data from the Italian power exchange, Gestore dei Mercati Energetici, for the period from January 1st, 2005, to October 31st, 2013, the study employed empirical modeling techniques such as testing for unit roots, checking for model specification, and addressing heteroskedasticity and serial correlation in residuals. Regression analysis revealed that an increase in the hourly average of daily production from wind power and solar energy led to a reduction in electricity prices. The results also noted an increase in the volatility of electricity prices with the increase in production from wind power and solar energy. Based on these findings, the study concludes that there is empirical evidence of the merit order effect in the Italian market during the observed period. Acknowledging and addressing the increase in electricity price volatility is essential for maintaining a functional energy market, as it poses challenges to energy security and affordability, key aspects of energy markets.

Cludius et al. (2014) studied the merit order effect in the German electricity markets, finding a significant reduction in spot market prices mainly due to increased electricity generation from wind and solar sources. They analyzed hourly electricity price data using tests like Augmented Dickey-Fuller and Phillips-Perron to check for unit roots. Using Ordinary Least Squares (OLS), they analyzed data from 2008 to 2012, focusing on wind power and solar energy. Their regression analysis examined how the spot price of electricity is influenced by wind power, solar energy, and total demand. The study highlights the benefits industries derive from lower wholesale electricity prices as renewable energy grows. The study also found that regulatory changes aimed at stabilizing wholesale prices often shift costs to consumers, disproportionately benefiting industries that can capitalize on the lower average of electricity.

The significance of investigating the influence of weather on electricity price dynamics, particularly in relation to the merit order effect, has been recognized within Sweden's context as well. A study by Macedo et al. (2021) researched the effect of wind power on the day-ahead electricity price dynamics in Sweden, considering the merit order effect. Utilizing hourly market data from Nord Pool spanning from January 1, 2016, to April 30, 2022, for the SE3 area, various variables such as electricity consumption prognosis, wind power prognosis, electricity inflow, and electricity outflow were examined. The study confirmed the Merit-Order Effect stemming from wind power, which consistently influenced prices throughout the day. The research

employed GARCH methodology to evaluate the impact of wind power generation on day-ahead electricity prices. The study also concluded that there were heightened volatility and price fluctuations linked to intermittent renewable energy sources.

2.2.2 Challenges with integrating of intermittent renewable energy and cross-border power connections

To achieve emission targets and transition to a sustainable energy system, the integration of renewable energy is of importance to ensure a reliable energy supply. Dispatchable conventional power plants, often utilizing fossil fuels, remain essential to the power system, especially during periods of low production from intermittent renewable energy sources. A study by Liebensteiner & Wrienz (2020) has shown empirical findings of the potential negative impact of intermittent renewable energy sources on investment in flexible thermal electricity generation capacity. The findings suggest a substantial economic impact, indicating a projected 20% decline in peak-load capacity within 5 years if renewable energy deployment mirrors recent trends. Despite aligning with EU targets to increase renewable energy share, the volatility in production necessitates flexible backup units, like gas-fired power plants, to mitigate adverse effects during unfavorable weather conditions. This underscores concerns that renewable energy may jeopardize long-term electricity supply.

Churkin et. al (2019) highlights challenges associated with cross-border power connections, including fluctuations in electricity prices and distrust stemming from regulatory and political barriers among countries. The study also highlighted the benefits of cross-border power connections, demonstrating the potential benefits of cooperation. These benefits include decreased generation costs, reduced consumer expenses, enhanced production efficiency, improved controllability, and mitigated climate impact. The study demonstrates how implementing concepts like the Cooperative Game Theory framework could potentially result in annual savings of \$7.1 billion if Northeast Asian countries engage in coordinated investment schemes. As a result, the construction of cross-border grids is considered a more efficient system in comparison to isolated grids, as it promotes enhanced regional collaboration and energy integration.

Interconnected electricity markets can also have negative impacts. For example, during the 2022 energy crisis, regions with low fossil fuel energy generation experienced increases in electricity prices when natural gas prices rose, as highlighted by Sveriges Riksbank (2023). Additionally, research on European electricity markets by Keles et al. (2020) emphasizes the correlation between Swiss, German, and French electricity prices, driven by factors such as gas prices and neighboring countries' electricity load. This illustrates how interconnected electricity markets can offer cost savings while also contributing to price volatility.

2.2.3 Future electricity market design

The Swedish Energy Research Center (Energiforsk), which coordinates energy research in Sweden, has initiated research programs to understand the evolving electricity market and its preparedness for future challenges. A paper by Brännlund et al. (2022) provides insightful conclusions and assessments regarding the necessity for reforms within the Swedish electricity market. Drawing from diverse research reports within the Energy Research Program FemD and other relevant studies, the discussion

highlights the anticipated surge in variability in future electricity supply and the heightened risks of potential inadequacies in power availability.

Sandberg (2024), also part of the research program FemD, examines the influence of Sweden's current wind power share (20%) on electricity prices. The study employed time series models to analyze the impact of wind speeds and temperature on electricity prices across four distinct price areas in Sweden: SE1, SE2, SE3, and SE4, from September 2022 to August 2023. The paper also examines which of the two variables - wind or temperature - better predicts electricity prices. The findings reveal a significant impact of both wind and temperature on electricity prices, with wind speeds exhibiting a more pronounced influence. However, initial observations suggest that there are additional factors beyond wind and temperature that contribute to electricity price fluctuations.

The literature emphasizes the critical aspect of future electricity market design and the challenges that lie ahead. The growth of renewable energy and the transition to sustainable energy systems present new opportunities but also pose challenges that must be addressed for the sustainable advancement of society.

2.3 Theoretical framework

2.3.1 Sustainable finance

Sustainable finance focuses on incorporating environmental, social, and governance (ESG) criteria into financial decision-making. It aims to promote economic growth while simultaneously reducing environmental impact and promoting social well-being. By aligning financial practices with sustainable development goals, corporations can address global issues such as climate change, resource depletion, and social inequality, all while pursuing organizational growth and commercial success. Sustainable finance includes financing renewable energy projects, investing in sustainable infrastructure, and supporting businesses that prioritize ESG principles. The value of sustainable finance lies in its potential to drive significant positive change by redirecting capital flows towards more sustainable activities. This not only helps mitigate environmental risks but also creates long-term economic opportunities and enhances societal well-being (Schoenmaker & Schramade, 2021).

Sustainable finance is also important for the transition to a sustainable energy future. This is done by channeling investments into renewable energy sources such as wind power, as well as supporting innovations in energy efficiency and storage technologies. The European Union describes sustainable finance as a tool for improving economic growth while reducing environmental impact. By financing projects that reduce reliance on fossil fuels and promote clean energy, sustainable finance helps accelerate the shift towards a low-carbon economy. This holistic approach not only supports the energy transition but also ensures that economic growth is achieved in a manner that benefits both people and the planet (Schoenmaker & Schramade, 2021).

2.3.2 Efficient market theory

The financial market movements are unpredictable, and efficient market theory emphasizes that constant new information is random and evolves the market and makes it more efficient. According to efficient market theory, "rational optimizers" are investors who can potentially increase the return on assets by making informed decisions. Fama et al. (1970) describe how prices adjust and respond to new

information, reflecting the market's evolving efficiency. The Efficient Market Hypothesis (EMH) explains that markets exist in three different forms, depending on the efficiency: weak, semi-strong, and strong. In the weak form, prices reflect all past market information, such as historical prices and volume. In the semi-strong form, prices incorporate all publicly available information, including financial statements and news releases. In the strong form, prices encompass all information, both public and private, meaning that even insider information is already reflected in the current prices.

Information asymmetry leads to market inefficiency, with the premises that the news/new details have not reached or been allocated accordingly to investors, details like for instance equity misvaluation (Li, 2020). According to efficient market theory, if markets are efficient, prices should adjust rapidly to reflect the new information as it becomes available, preventing consistent arbitrage opportunities. However, the unique characteristics of different markets can complicate full application of the efficient market theory (Fama, 1970).

2.3.3 Market efficiency in electricity pricing

Building on the principles of the Efficient Market Theory proposed by Fama et al. (1970), it is argued that markets operate efficiently when prices accurately reflect all available information. When applied to electricity markets, this theory suggests that electricity prices should encompass all factors influencing them for the market to be considered efficient (Joskow, 2006)). Therefore, in an efficient electricity market, prices should adjust to incorporate any new information as it emerges, effectively preventing consistent arbitrage opportunities.

Electricity markets typically operate within regulated or deregulated frameworks. Regulated markets, while prioritizing stability and consumer protection, can sometimes inadvertently suppress competition and hinder innovation (Bessembinder & Lemmon, 2002). Deregulated electricity markets operate under market forces such as supply and demand, fostering competition that has the potential to reduce electricity prices for consumers. In deregulated electricity markets, according to the Efficient Market Theory, prices are expected to reflect all relevant factors influencing electricity prices for the market to be deemed efficient. Any deviation from this expectation could raise questions about the theory's validity when applied to electricity markets.

In the electricity market, financial risk can be attributed to various factors, including weather conditions, particularly affecting renewable energy sources. Weather introduces uncertainty, impacting electricity market dynamics, leading to revenue fluctuations, investment risks, and operational disruptions. For example, changes in wind conditions significantly affect wind turbine performance and subsequently electricity production capacity. Furthermore, weather patterns influence supply chain dynamics, spot market operations, and bid prices, contributing to unforeseen market swings and spot price changes that affect consumers (Londoño & Velásquez, 2023)

2.3.4 The electricity market

Despite Sweden's minimal reliance on fossil fuels for electricity generation, its integration into the broader European electricity market suggests that a correlation with fossil fuel prices remains relevant. This is particularly evident due to the reliance of other European countries, such as Germany and Poland, on fossil fuels, with coal constituting a significant portion of their electricity generation mix. For instance,

Denmark, Germany and Poland, key electricity trading partners with Sweden, have 13 %, 33% and 70% of coal in their respective total electricity generation mixes (IEA, 2023). While Sweden itself relies heavily on renewable energy sources, operating within a deregulated framework, both Sweden and the larger European electricity market are subject to the impacts of fossil fuel prices, despite disparities in their respective electricity generation capacity. This underscores the importance of the integrated EU electricity market in ensuring secure, sustainable, and affordable energy supplies for EU citizens. Figure 3 illustrates the electricity mix of Sweden, Denmark, Germany, and Poland in 2023 by electricity generation source.

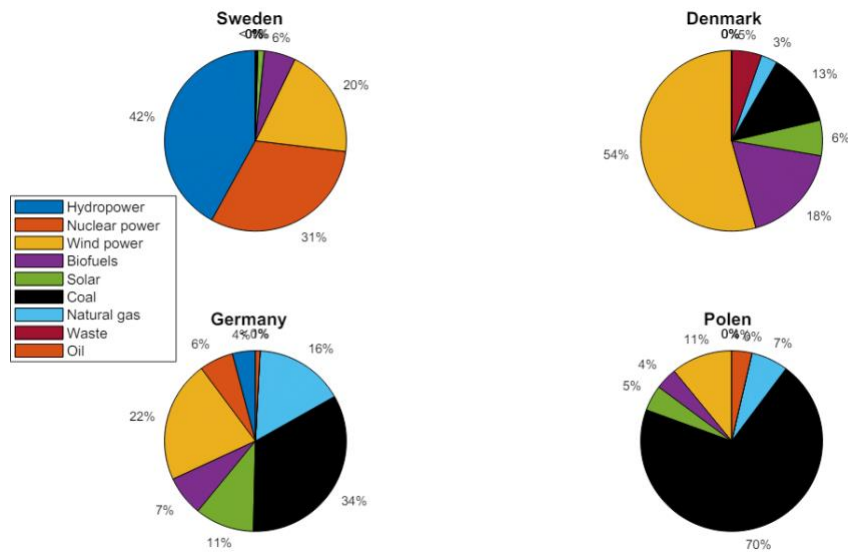


Figure 3. The electricity mix of Sweden, Denmark, Germany, and Poland (IEA, 2023).

The price of electricity is determined by the last generator dispatched to meet demand within each time interval in the locational pricing setup (Stoft, 2002)). This means that factors such as fuel costs, demand levels, and the availability of renewable energy sources like wind and solar power influence electricity prices. Typically, cheaper, and more efficient generators, such as those using renewable energy sources, are dispatched first, followed by more expensive options like natural gas or coal-fired power facilities (Creti & Fontini, 2019). Power generation facilities submit bids indicating the price at which they are willing to supply electricity to the grid, and these bids are then ranked based on the cost per unit of electricity generated. This system ensures that electricity is produced at the lowest possible cost to consumers, a concept known as the merit order effect. Figure 5 illustrates the concept of the merit order effect, where power sources are ranked based on their marginal cost. The spot price is determined by the power sources with the highest marginal cost of producing electricity, in this case natural gas.

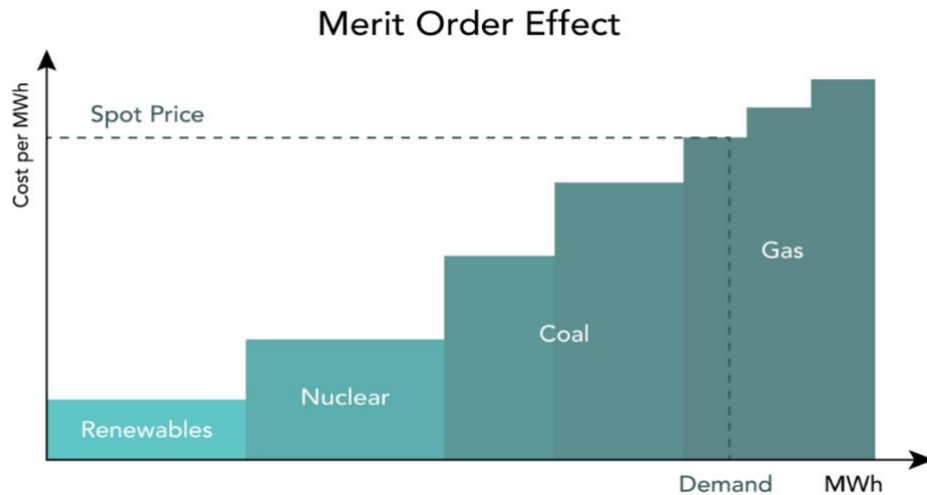


Figure 4. Illustration of the merit order effect in electricity markets (Squeaky Energy, 2023).

Electricity market trading in Sweden is structured around different time intervals, each defining a distinct market type. At the furthest end from the delivery date lies the forward electricity market. The forward market enables securing electricity prices years in advance as a risk mitigation strategy. Traders often utilize such contracts for their low volatility characteristics, making it an attractive option for long-term planning and hedging purposes (Svenska Kraftnät, 2023).

The spot market, the largest electricity market by trading volume, operates between 36 hours and 12 hours before actual electricity delivery. It represents a key market for short to medium-term electricity trading, enabling market participants to respond to near-term demand and supply dynamics with flexibility.

The intraday market operates up to one hour before delivery. This market allows for flexibility to adjust to real-time conditions in response to unforeseen developments or changes in market conditions.

The balancing market operates during the delivery period and primarily involves grid operators who utilize it to regulate grid frequency and ensure system stability. This market is the final trading opportunity for electricity and is key for real-time adjustments to maintain grid reliability and operational integrity.

2.4 Conclusion

In conclusion, we observe from the literature that the growth of renewable energy has significantly influenced the electricity markets. The merit order effect, highlighted by previous research, indicates a general decrease in the cost of electricity due to the lower marginal cost of electricity generation from renewable energy. Volatility in the electricity market increases because renewable energy sources like wind and solar cannot consistently generate power like other common electricity generation sources such as hydropower or nuclear power. Furthermore, we note a high correlation among markets engaged in electricity trading, despite variations in the presence of fossil fuel generation. This correlation is driven by trading mechanisms that affect the broader energy landscape. Regulatory frameworks and physical constraints within the electricity grid also emerge as crucial factors influencing market efficiency.

3. Research design

3.1 Problem, purpose, and contribution

The study aims to empirically analyze how weather variables and fossil fuel prices relate to electricity prices in Sweden. We seek to understand the significance of these factors' impact on the country's electricity markets (SE1, SE2, SE3 and SE4), recognizing that empirical analysis utilizing real time market data provides the most reliable answers to such questions. The study employs regression analysis with electricity prices as the dependent variable. It includes weather variables (wind speed and temperature), fossil fuel prices (natural gas and coal), and control variables (seasons and the GDP gap) to assess their impact on Sweden's electricity markets from 2013 to 2023. The knowledge contribution of this research is to demonstrate how these factors have influenced the Swedish electricity market and how these impacts vary across different electricity price areas within the country. The following hypotheses are examined.

Hypothesis 1

H₀: There is no significant influence of weather variables (wind speeds and temperature) on electricity prices.

H₁: Weather variables (wind speeds and temperature) significantly influence electricity prices.

Hypothesis 2

H₀: There is no significant influence of fossil fuel prices (natural gas and coal) on electricity prices.

H₁: Fossil fuel prices (natural gas and coal) significantly influence electricity prices.

3.2 Scientific perspective

This thesis adopts a quantitative, deductive, and positivistic scientific perspective. Quantitative research seeks to test theories using numerical data and maintaining an objective stance, as explained by Bryman & Bell (2005). The deductive approach used involves formulating and testing a hypothesis based on an existing general theory, beginning with a theoretical framework that is evaluated through real-world data (Bryman & Bell, 2017). Therefore, this thesis tests the broad hypothesis that weather patterns and energy commodities affect the electricity market, utilizing data from sources like NordPool, Swedish Meteorological and Hydrological Institute, the Swedish National Institute of Economic Research, and the International Monetary Fund. Thus, quantitative research is appropriate here, as it allows for the empirical testing of a pre-established theory while ensuring an objective perspective (Bryman & Bell, 2005).

3.3 Data

Access to the Nord Pool electricity exchange database facilitates the exploration of price fluctuations across European markets, including Sweden, Norway, Denmark, and Finland. In this study, historical hourly electricity spot prices from 2013 to 2023 were obtained from Nord Pool for all Swedish electricity markets, namely SE1, SE2, SE3, and SE4. The data was provided in an Excel format, quoted in Swedish kronor per megawatt-hour (SEK/MWh).

Given the empirical focus on understanding the relationship between weather variables, fossil fuel prices, and electricity prices in Sweden, the study primarily examines weather data and energy commodity prices. Weather data, sourced from the Swedish Meteorological and Hydrological Institute, included hourly measurements from weather stations located in major cities of each price area: Luleå-Kalix Airport, Sundsvall-Timrå Airport, Stockholm-Bromma Airport, and Malmö Airport. The weather variables considered in the analysis included wind speeds and temperature measurements, recorded on an hourly basis. Energy commodity price data were sourced from the IMF Primary Commodities Database. The analysis includes energy commodities such as natural gas and coal. Price data, measured monthly and quoted in US dollars, was provided in an Excel sheet. Exchange rates provided by Sveriges Riksbank were utilized to convert the prices to Swedish kronor.

To address potential influences on electricity prices beyond the main focus of our analysis, we introduce control variables. These control variables help mitigate the effects of external factors on the dependent variable, ensuring a clearer examination of the relationship between the independent variables of interest. Specifically, control variables for economic conditions in Sweden (GDP gap) and seasonality are included in the study. The data for GDP gap data was collected from Konjunkturinstitutet and measured as an annual percentage.

Table 1. Overview of variables used in analysis.

Variable	Description	Expected sign	Data source
Electricity prices _{x,t}	The electricity spot price is measured in swedish kronor per megawatt hour (SEK/MWh) for electricity price area (x) at time (t)	Dependent variable	Nord Pool
Wind speeds _{x,t}	Wind speeds are measured in meters per second (m/s) for weather station (x) at time (t)	-	SMHI
Temperature _{x,t}	The temperature is measured in degrees Celsius (°C) for weather station (x) at time (t)	-	SMHI
Natural gas prices	Natural gas prices represent the prices of natural gas (Dutch TTF Natural Gas Futures) quoted in US dollars per million metric British thermal units.	+	IMF
Coal prices	Coal prices represent the prices of South African export coal quoted in US dollars per metric ton.	+	IMF
Season	Dummy variables for the season of the year, either winter, spring, summer or autumn.		
GDP Gap	The economic state of Swedish economy defined by Konjunkturinstitutet as economic boom or downturn	+	KI

3.4 Method

The primary analytical approach employed in this study is multiple linear regression analysis based on the ordinary least square's method. The regression model utilized in this study is formulated as follows:

$$Y = \beta_0 + \beta_1 + \beta_2 + \beta_3 + \beta_4 + \beta_5 + \beta_6 + \varepsilon$$

Where,

- Y = the dependent variable, which is the electricity spot price measured in Swedish kronor per megawatt-hour (SEK/MWh) for electricity price area x at time t .
- β_0 = the intercept
- β_1 = the wind speeds measured in meters per second (m/s) for the weather station x at time t .
- β_2 = the temperature measured in degrees Celsius ($^{\circ}\text{C}$) for the weather station x at time t .
- β_3 = the prices of natural gas (Dutch TTF Natural Gas Futures) quoted in US dollars per million metric British thermal units.
- β_4 = the prices of South African export coal, quoted in US dollars per metric ton.
- β_5 = the economic state of the Swedish economy defined by Konjunkturinstitutet as an economic boom or downturn.
- β_6 = variables for the season of the year, categorized as winter, spring, summer, or autumn.
- ε = the error term

The main reasons for employing linear regression stem from its wide-ranging applicability in transforming raw data into actionable insights and business intelligence. Coefficients in the regression model measure the strength and direction of the relationship between the independent and dependent variables, providing valuable insights into their associations. Since the regression model encompasses variables with different units, such as meters per second for wind speeds and degrees Celsius for temperature, standardization of all variables was conducted. Standardization ensures that each variable contributes evenly to the output of the regression model, thereby facilitating the interpretation of results and enabling meaningful conclusions to be drawn from the raw data (Brooks, 2019; Campbell, Lo, & MacKinlay, 1996).

The model incorporates standard assumptions of the linear regression model, which is based on four assumptions to ensure the validity and reliability of its results. These assumptions include linearity, homoscedasticity, independence, and normality. Linearity ensures a linear relationship between x and y . Homoscedasticity implies that the variance of the residuals is constant for any value of x . Independence states that the observations are independent of each other. Normality assumes that for any fixed value of x , y is normally distributed. As the thesis is a bachelor's level thesis in business administration, it focuses on simpler empirical analyses, preferring straightforward methods over advanced techniques. Acknowledging these four assumptions, the thesis operates under the premise that they hold for the sample. Despite this, it's crucial to recognize that these assumptions may not always apply in real-world scenarios,

emphasizing the necessity of thorough residual analysis and testing to ensure the model's reliability and validity (Brooks, 2019; Campbell, Lo , & MacKinlay, 1996).

To understand the correlation between the dataset, we utilize the Pearson correlation coefficient. The Pearson correlation coefficient ranges from a value of -1 to 1, where 1 indicates a perfect positive correlation, -1 a perfect negative correlation and 0 no correlation between the datasets (Brooks, 2019; Campbell, Lo , & MacKinlay, 1996).

To assess the predictive capability of the regression model for the observed electricity spot price, we use the coefficient of determination (R^2) to interpret how well the independent variables predict the variation in the dependent variable. The coefficient of determination ranges between 0 and 1 and is calculated for each price area and respective regression model. It is calculated by dividing the sum of squared regression (SSR), also known as the variation explained by the regression model, by the total variation of data, as the sum of squared total (SST) (Brooks, 2019; Campbell, Lo , & MacKinlay, 1996).

To test the null hypothesis and determine whether to reject it or not, p-values serve as a crucial measure. A p-value indicates the likelihood of obtaining observed test results assuming the null hypothesis is true. By examining p-values at different significance levels (e.g., 10%, 5%, and 1%), we gain an understanding of each variable's statistical significance. Additionally, confidence intervals aid in estimating the probable range of the true population parameter. The application of linear regression in energy markets sheds light on how independent variables influence electricity prices, thereby empowering decision-makers in the energy sector (Brooks, 2019; Campbell, Lo , & MacKinlay, 1996).

3.5 Source critical considerations

The thesis is conducted based on information derived from academic journals and peer-reviewed articles. The articles have undergone quality control to ensure credibility, with information verified through verification process and sources are carefully referenced. For our study, we retrieved historical data from various credible and publicly accessible platforms, including Nord Pool, the Swedish Meteorological and Hydrological Institute, the International Monetary Fund, and the Swedish National Institute of Economic Research. Nord Pool, also known as the Nordic electricity exchange, is a leading source of electricity price data in Europe. All these sources contributed to our research study and helped us address our research question.

3.6 Research ethical reflection

The research focuses on quantitative data analysis and involves no direct interaction with human subjects which diminishes ethical concerns typically associated with personal data and direct human impact, such as confidentiality and anonymity. However, it still requires rigorous attention to ethical considerations around large-scale data sets it utilizes, particularly regarding data integrity. The research mandates transparent reporting on how data was collected, processed, and analyzed.

3.7 Validity and reliability of the study

Ensuring the validity and reliability of our research is crucial. Validity refers to the accuracy of our findings and whether the research truly measures what it intends to

measure. To enhance validity, we used data from reputable sources and employed standardized methods for data collection and analysis. Reliability, on the other hand, focuses on the consistency of our results. By using established methodologies and replicable procedures, we aim to achieve consistent outcomes across different studies and over time. These measures ensure that our findings are both credible and dependable.

4. Analysis and findings

4.1 Descriptive statistics and correlation

This section presents a summary of descriptive statistics for the sample considered in the regression model. Table 2 shows the summary presented in the mean value, median value, standard deviation, minimum and maximum value.

The maximum hourly electricity spot price is 6432.42 SEK/MWh in SE1 and SE2, and 8513.26 SEK/MWh in SE3 and SE4. The corresponding minimum hourly electricity spot prices was -690.49 SEK/MWh for all electricity price areas. The mean, median, and standard deviation of electricity prices are lowest in SE1 (356.52, 299.84, 334.41) and highest in SE4 (526.04, 335.91, 663.26). The lower standard deviation in SE1 indicates greater price consistency compared to the more volatile electricity market of SE4, which experiences greater fluctuations.

The highest electricity spot price is approximately 9.3 times greater than the lowest in SE1 and SE2, and about 12.3 times greater in SE3 and SE4. In contrast, natural gas prices can be up to 48 times higher at their peak, and coal prices up to 6.7 times higher. Since weather variables remain relatively constant year to year, describing them does not significantly contribute to understanding the electricity market. However, the correlation between weather variables and electricity prices, as well as energy commodities, is interesting to analyze.

Table 2. Descriptive statistics.

Variable	Mean	Median	Std.Dev	Min	Max
Electricity spot price (kr/MWh_SE1)	356.52	299.84	334.41	-690.49	6432.43
Electricity spot price (kr/MWh_SE2)	359.34	300.46	339.95	-690.49	6432.43
Electricity spot price (kr/MWh_SE3)	466.91	317.75	584.57	-690.49	8513.26
Electricity spot price (kr/MWh_SE4)	526.04	335.91	663.26	-690.49	8513.26
Wind speeds (m/s_SE1)	3.31	3.00	2.17	0	17.00
Wind speeds (m/s_SE2)	3.00	3.00	2.07	0	16.00
Wind speeds (m/s_SE3)	3.26	3.00	1.95	0	16.00
Wind speeds (m/s_SE4)	2.91	2.70	1.87	0	16.00
Temperature (°C_SE1)	3.35	3.10	10.20	-36.40	30.30
Temperature (°C_SE2)	4.35	3.60	9.34	-28.2	31.00
Temperature (°C_SE3)	7.13	6.50	8.24	-21.90	32.40
Temperature (°C_SE4)	9.72	9.30	7.16	15.40	31.60
Natural gas prices (kr/MMBTU)	116.73	80.88	113.01	15.52	746.65
Coal prices(kr/MT)	1071.16	877.88	660.92	528.78	3532.41

Table 3-6 shows the correlation coefficients between electricity spot price, wind speeds, natural gas prices, and coal prices, categorized by price area. Wind speeds generally show a negative correlation with electricity prices in all areas. This indicates that higher wind speeds, which lead to increased wind power generation, tend to reduce electricity spot prices, showing an inverse correlation. We observe an increasing negative correlation from SE1 (-0.080) to the southernmost price region, SE4 (-0.146). This can be explained by the proximity to Denmark and Germany, which have high wind power production (54% and 20%, respectively) and cross-border connections with SE4. These

market dynamics suggest a higher negative correlation between wind speeds and electricity prices in SE4 compared to the more northern price areas.

Temperature exhibits a negative correlation with electricity prices in all areas, ranging from -0.018 in SE4 to -0.136 in SE1. This indicates that increases in temperature, which often decrease electricity demand for heating, have an inverse relationship with electricity prices. The highest negative correlation is observed in SE1 (-0.136), suggesting that temperature variations have a slightly more pronounced impact on electricity prices in this region.

Natural gas prices exhibit a positive correlation with electricity prices across all areas, with the strongest correlations in SE3 (0.585) and SE4 (0.635). This relationship indicates that rising natural gas prices lead to higher electricity prices. Coal prices similarly show a positive correlation with electricity prices, especially in SE3 (0.529) and SE4 (0.593), highlighting the influence of coal on electricity generation costs. This supports our previous assertion that Poland and Germany, being major coal and natural gas producers, influence SE4 due to cross-border transmission cables, thereby impacting Swedish electricity spot prices due to market integration.

Table 3. Correlation matrix for price area SE1 Luleå.

	Electricity spot price (SE1)	Wind speeds (SE1)	Temperature (SE1)	Natural gas prices	Coal prices
Electricity spot price (SE1)	1.000				
Wind speeds (SE1)	-0.080	1.000			
Temperature (SE1)	-0.136	-0.023	1.000		
Natural gas prices	0.266	-0.073	0.007	1.000	
Coal prices	0.262	-0.080	0.027	0.891	1.000

Table 4. Correlation matrix for price area SE2 Sundvall.

	Electricity spot price (SE2)	Wind speeds (SE2)	Temperature (SE2)	Natural gas prices	Coal prices
Electricity spot price (SE2)	1.000				
Wind speeds (SE2)	-0.074	1.000			
Temperature (SE2)	-0.124	0.033	1.000		
Natural gas prices	0.304	-0.037	0.003	1.000	
Coal prices	0.287	-0.037	0.020	0.891	1.000

Table 5. Correlation matrix for price area SE3 Stockholm.

	Electricity spot price (SE3)	Wind speeds (SE3)	Temperature (SE3)	Natural gas prices	Coal prices
Electricity spot price (SE3)	1.000				
Wind speeds (SE3)	-0.129	1.000			
Temperature (SE3)	-0.116	0.121	1.000		
Natural gas prices	0.585	-0.054	-0.013	1.000	
Coal prices	0.529	-0.040	0.007	0.891	1.000

Table 6. Correlation matrix for price area SE4.

	Electricity spot price (SE4)	Wind speeds (SE4)	Temperature (SE4)	Natural gas prices	Coal prices
Electricity spot price (SE4)	1.000				
Wind speeds (SE4)	-0.146	1.000			
Temperature (SE4)	-0.018	-0.023	1.000		
Natural gas prices	0.635	-0.073	0,007	1,000	
Coal prices	0.593	-0.080	0,027	0,891	1,000

4.2 Regression model

This section presents the results of the regression models, providing insights into the relationships between the studied independent variables and electricity prices. Table 7-10 shows results of the regression for all electricity price areas. We see that the R-squared value was similar to the adjusted r-square value for the first three decimals in the tables presented below, but differ on the fifth decimal, hence why it looks similar. The study only presents results for three decimals.

In electricity price area SE1, regression (1), which includes only the wind speed measured at the Luleå-Kalix Airport variables, yields a beta coefficient of -0.080 and a standard error of 0.003.

In table 7, which shows the regression results for SE1, we see that wind speed has a negative coefficient across all regression models. This indicates that an increase in wind speed tends to reduce electricity prices. The beta coefficient of wind speed increases and decreases for every new variable added. Temperature also shows a negative coefficient in all models, suggesting that higher temperature leads to lower electricity prices. Coal and natural gas on the other hand, shows positive coefficients which indicates that higher coal and natural gas prices leads to higher electricity prices.

The R-squared value of wind speed is 0.006 in (1). In regression (2), which includes temperature, the R-squared value increases to 0.022. With all weather variables (wind speeds and temperature) included in regression (3), the R-squared value reaches 0.094. This indicates that while the control variables and fossil fuel prices have significant explanatory power, the weather variables add only marginal explanatory power. In regression (4), the coefficient for wind speeds is -0.064.

On table 7 (4), we observe that when all variables are added, the beta coefficient for natural gas is highest. Furthermore, it increases in value, while the others decrease. This indicates that natural gas has the strongest correlation with the electricity spot price. We see that all the variables are significant for all the regression models at 0.01, indicating that all the variables have a significant influence on electricity prices in the electricity price area SE1 Luleå.

Table 7. Regression results for electricity price area SE1 Luleå.

Dependent variable	(1)	(2)	(3)	(4)
Wind speeds	-0.080 *** (0.003)	-0.083 *** (0.003)	-0.062 (0.003)	-0.064 *** (0.003)
Temperature		-0.126 *** (0.003)	-0.129 (0.003)	-0.091 *** (0.005)
Natural gas			0.145 *** (0.007)	0.173 *** (0.007)
Coal			0.132 *** (0.007)	0.052 *** (0.007)
Summer				-0.071 *** (0.006)
Fall				-0.040 *** (0.004)
Spring				-0.107 *** (0.004)
GDP gap				0.125 *** (0.003)
Observations	96408	96408	96408	96408
R square	0.006	0.022	0.094	0.114
Adjusted R Square	0.006	0.022	0.094	0.114

Standard errors are given in parenthesis. *** p<0.01, ** p<0.05, * p<0.10

In the electricity price area SE2 Sundsvall, regression (1) focuses solely on wind speed measured at the Sundsvall-Timrå Airport, yielding a beta coefficient of -0.074 with a standard error of 0.003.

Table 8, illustrating the regression outcomes for SE2, consistently portrays wind speed with a negative coefficient across all models. This indicates that an increase in wind speed tends to lower electricity prices. Moreover, the beta coefficient of wind speed fluctuates with the inclusion of each new variable. Temperature also consistently exhibits a negative coefficient in all models, suggesting that higher temperatures lead to lower electricity prices. In contrast, both coal and natural gas display positive coefficients, implying that higher prices for these commodities correlate with higher electricity prices.

The R-squared value for wind speed stands at 0.006 in regression (1). Upon the incorporation of temperature in regression (2), the R-squared value increases to 0.023. When all weather variables (wind speeds and temperature) are included in regression

(3), the R-squared value further rises to 0.116. This indicates that while control variables and fossil fuel prices offer significant explanatory power, the contribution of weather variables remains marginal.

In regression (4), the coefficient for wind speeds is -0.056. Table 7 (4) illustrates that with all variables included, the beta coefficient for natural gas emerges as the highest. Furthermore, it demonstrates an increasing trend in value, while the coefficients for other variables decrease. This suggests that natural gas holds the strongest correlation with the electricity spot price.

Importantly, all variables maintain statistical significance across all regression models at the 0.01 confidence level, highlighting their substantial influence on electricity prices in the electricity price area SE2, Sundsvall.

Table 8. Regression results for electricity price area SE2.

Dependent variable	(1)	(2)	(3)	(4)
Wind speeds	-0.074 *** (0.003)	-0.070 *** (0.003)	-0.058 *** (0.003)	-0.056 *** (0.003)
Temperature		-0.134 *** (0.003)	-0.136 *** (0.003)	-0.127 *** (0.005)
Natural gas			0.223 *** (0.007)	0.257 *** (0.007)
Coal			0.089 *** (0.007)	0.005 *** (0.007)
Summer				-0.012 *** (0.005)
Fall				-0.022 *** (0.004)
Spring				-0.078 *** (0.004)
GDP gap				0.126 *** (0.003)
Observations	96408	96408	96408	96408
R square	0.006	0.023	0.116	0.134
Adjusted R Square	0.006	0.023	0.116	0.134

Standard errors are given in parenthesis. *** p<0.01, ** p<0.05, * p<0.10

In the regression results in Table 9 for the electricity price area SE3, wind speeds show a negative coefficient across all models. This indicates that higher wind speeds help to lower electricity prices. Temperature also consistently shows a negative coefficient across all models, suggesting that higher temperatures correlate with lower electricity prices. This effect is consistent and significant, underlining temperature as a critical factor in electricity demand and pricing in Stockholm.

For SE3, the R-squared value for regression (1), including only the wind speed, is 0.017. In regression (2), with the inclusion of temperature, the R-squared value is 0.027. Regression (3), R-square value is 0.361, natural gas having the highest beta coefficient 0.534 and correlation and temperature being the lowest -0.099. In regression (4), the R-squared value further increases to 0.374, the highest among the three areas. This indicates that natural gas and coal prices have the largest explanatory power for SE3, followed by control variables, with weather variables having the least impact.

The influence of natural gas prices is significant in Regression (3) and (4), indicating that when natural gas prices increase, electricity prices also tend to rise. Coal shows consistent positive coefficient which means that a higher price on coal also increases electricity prices. The models themselves, as shown by the R-squared values, indicate increasing explanatory power as more variables are included, with model 4 explaining the highest proportion of variability in electricity prices. This suggests that the later models, which likely include more comprehensive data on energy sources and possibly other economic indicators, provide a better fit for understanding electricity price fluctuations in Stockholm.

Table 9. Regression results for electricity price area SE3 Stockholm.

Dependent variable	(1)	(2)	(3)	(4)
Wind speeds	-0.129 *** (0.003)	-0.116 *** (0.003)	-0.086 *** (0.003)	-0.080 *** (0.003)
Temperature		-0.101 *** (0.003)	-0.099 *** (0.003)	-0.105 *** (0.004)
Natural gas			0.534 *** (0.006)	0.573 *** (0.006)
Coal			0.051 *** (0.006)	-0.035 *** (0.006)
Summer				0.009 ** (0.005)
Fall				0.005 (0.003)
Spring				-0.039 *** (0.003)
GDP gap				0.117 *** (0.003)
Observations	96408	96408	96408	96408
R square	0.017	0.027	0.361	0.374
Adjusted R Square	0.017	0.027	0.361	0.374

Standard errors are given in parenthesis. *** p<0.01, ** p<0.05, * p<0.10

The regression results displayed in Table 10 for electricity price area SE4 shows that wind speeds show a significantly negative coefficient across all models, indicating that higher wind speeds result in lower electricity prices. This effect is consistent and substantial, suggesting a stronger dependency of the electricity market on wind energy

in this area. Temperature has a negative effect on electricity prices in the initial models, indicating that higher temperatures, likely reducing heating demands, correlate with lower electricity prices. However, the influence of temperature diminishes in model (4), which could suggest that other variables, when controlled, mitigate the impact of temperature on electricity prices.

Natural gas prices show a strong positive correlation with electricity prices across the models where this variable is included. The significance of these results underscores natural gas as a critical factor in electricity price determination, likely due to its role in power generation in SE4. Coal, similar to natural gas, exhibits a positive influence on electricity prices in the initial models but this influence diminishes in the later models, indicating that coal's impact might be contingent on the presence of other factors in the model.

In SE4, regression (1) the R-square value for wind speed in relation to the electricity spot price is 0.21. In regression (2) with two variables, wind and temperature the R-square value increases to 0.022, the standard error remains the same 0,003. In regression (3) the R-square is 0,417, with both the weather variables and commodities included it is evident that natural gas has the highest beta coefficient 0.515. Regression (4) the control variables are included and the R-square is 0.427.

The regression models progressively include more variables, as reflected in the increasing R-squared values, particularly a notable jump in the later models. This improvement in R-squared values indicates a substantial enhancement in the model's ability to explain the variations in electricity prices when incorporating a broader set of variables.

Table 10. Regression results for electricity price area SE4 Malmö.

Dependent variable	(1)	(2)	(3)	(4)
Wind speeds	-0.146 *** (0.003)	-0.146 *** (0.003)	-0.099 *** (0.002)	-0.100 *** (0.003)
Temperature		-0.022 *** (0.003)	-0.027 *** (0.002)	-0.006 (0.004)
Natural gas			0.515 *** (0.005)	0.551 *** (0.006)
Coal			0.127 *** (0.005)	0.050 *** (0.006)
Summer				-0.032 *** (0.004)
Fall				-0.031 *** (0.003)
Spring				-0.044 *** (0.003)
GDP gap				0.103 *** (0.003)
Observations	96408	96408	96408	96408
R square	0.021	0.022	0.417	0.427
Adjusted R Square	0.021	0.022	0.417	0.427

Standard errors are given in parenthesis. *** p<0.01, ** p<0.05, * p<0.10

To enhance readability, table 11 summarizes all regression (4) for the price areas studied.

Table 10. Regression results for electricity price area SE4.

Dependent variable	SE1 (4)	SE2 (4)	SE3 (4)	SE4 (4)
Wind speeds	-0.064 *** (0.003)	-0.056 *** (0.003)	-0.080 *** (0.003)	-0.100 *** (0.003)
Temperature	-0.091 *** (0.005)	-0.127 *** (0.005)	-0.105 *** (0.004)	-0.006 (0.004)
Natural gas	0.173 *** (0.007)	0.257 *** (0.007)	0.573 *** (0.006)	0.551 *** (0.006)
Coal	0.052 *** (0.007)	0.005 *** (0.007)	-0.035 *** (0.006)	0.050 *** (0.006)
Summer	-0.071 *** (0.006)	-0.012 *** (0.005)	0.009 ** (0.005)	-0.032 *** (0.004)
Fall	-0.040 *** (0.004)	-0.022 *** (0.004)	0.005 (0.003)	-0.031 *** (0.003)
Spring	-0.107 *** (0.004)	-0.078 *** (0.004)	-0.039 *** (0.003)	-0.044 *** (0.003)
GDP gap	0.125 *** (0.003)	0.126 *** (0.003)	0.117 *** (0.003)	0.103 *** (0.003)
Observations	96408	96408	96408	96408
R square	0.114	0.134	0.374	0.427
Adjusted R Square	0.114	0.134	0.374	0.427

Standard errors are given in parenthesis. *** p<0.01, ** p<0.05, * p<0.10

4.3 Annual development of beta coefficients for weather variables

To understand the annual standardized beta coefficient development for all the weather variables, we conducted regressions on annual data and plotted the lines on a graph to visually interpret any trends observed from 2013 to 2023. This plot allows us to determine if there is a negative trend for the weather variables, indicating an increasing negative influence on electricity prices over the studied period (see Figure 1). From the plot, we observe smoother lines for electricity price areas SE4 and SE2 for wind speeds, although some years exhibit peaks. The price area that shows the least resemblance to a negative trend is SE3. This discrepancy can be attributed to the fact that wind speeds from the measured weather stations may not accurately reflect the power generation capacity in this electricity price area.

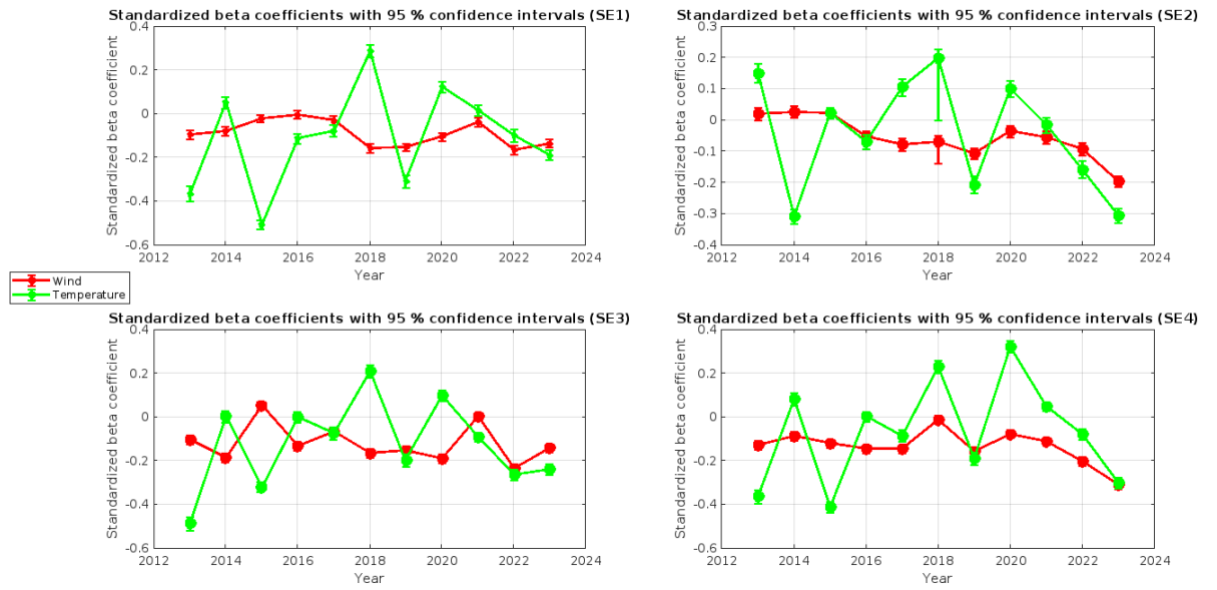


Figure 5. Annual development of beta coefficients for weather variables.

5. Discussion

The study has identified an empirical relationship between weather variables and fossil fuel prices, rejecting the null hypothesis and addressing the first two research questions of the thesis. We observe that the electricity price region significantly influences how independent variables affect electricity prices. This finding aligns with previous research on integrated energy markets, which emphasizes the interconnected nature of energy systems and their varying impacts on electricity prices (Churkin, 2019).

In particular, the SE4 electricity price area, covering Malmö and connected to Germany, Poland, Denmark, and Lithuania via cross-border power connections, consistently displayed the highest R-squared value across different regression models. This high explanatory power can be attributed to two primary factors. Firstly, the proximity to electricity markets with significant fossil fuel generation affects electricity pricing in SE4, resulting in a stronger correlation with fossil fuel prices. Secondly, the closeness to Denmark, which has substantial wind power generation capacity, links wind speeds to electricity prices, thereby impacting the dynamics of SE4.

From our analysis, we concluded that all price areas, except SE4, exhibited a higher beta coefficient for temperature than for wind speeds. These outcomes are consistent with Sandberg's (2024) findings that wind speed impacts the electricity spot price more significantly than temperature, with rising wind speeds contributing to this effect. This supports a broader trend in the Swedish electricity market and other markets with high levels of renewable energy. Historically, electricity markets in cold winter regions have shown a significant price correlation with temperature. However, as society shifts towards sustainable development and prioritizes renewable energy, this historical correlation is challenged, particularly in SE4. The high concentration of wind power in SE4 and surrounding areas has significantly altered the historical correlation between weather variables and electricity prices.

The merit order effect, and its impact on lowering electricity costs while simultaneously increasing volatility, as found by researcher (Clò, Cataldi, & Zoppoli, 2015; Cludius, Hermann, Matthes, & Graichen, 2014; Macedo, Marques, & Damette, 2021) is evident in our report through the negative coefficient for wind speeds in relation to electricity prices. This corroborates the fact that higher wind speeds reduce the average cost of electricity in the market. While volatility was not the primary focus of our study, we found that the volatility from the energy crisis in Europe in 2022 was more impactful on the electricity market. This finding supports the study by Liebensteiner, which discusses the necessity of flexible thermal electricity generation capacity to mitigate adverse effects during unfavorable weather conditions. In our case, we see that the volatility was more attributed to the energy crisis, where a general decline in affordable thermal electricity generation capacity, particularly in the form of natural gas, experienced significant price increases, jeopardizing the electricity supply.

Hagfors (2016) mentions using probability models as a risk management tool utilized by market participants to accurately forecast prices with assurance. For a continued growth of the sustainable energy system, and attracting feasible investment opportunities, a tool for managing risk with weather derivatives is necessary for continued expansion of intermittent renewable energy in the absence of large scale energy storage. According to the CME 30 % of the US economy is directly affected by the weather (CME,2020). For that reason there is a real need to hedge the risk they face

from weather conditions negatively impacting them. There is no such system in Sweden. For continued expansion of renewable energy and transition to a sustainable energy system, it is important to ensure functional capital markets and provide incentives for investors by limiting losses from weather-related risks.

The research is grounded in the paradigm shift towards sustainable energy systems since 2010 and aims to understand the impact of weather on energy markets. It diverges from previous research conducted before 2010, which primarily focused on extreme weather's influence on energy assets, as discussed and demonstrated by the author Mu. Looking ahead, we anticipate that research will continue along this trajectory, aiming to comprehend the continuous impact of weather on energy markets rather than solely focusing on extreme weather events. This expectation arises from the ongoing expansion and development of weather-dependent sustainable energy sources within society.

Understanding this impact is crucial for adequately modeling the financial performance of investments in sustainable energy systems. Given the perceived volatility in renewable energy and the uncertainty of weather patterns, the report also acknowledges the potential jeopardization of long-term electricity supply due to the proliferation of renewable energy sources. Historically, electricity systems have relied on dispatchable electricity sources, meaning they can be brought online based on demand. However, the introduction of supply variability presents a novel challenge in energy markets. This is corroborated by Liebensteiner et. al, who found empirical evidence of the potential negative impact of intermittent renewable energy sources on investment in flexible thermal electricity generation capacity.

To manage this challenge, adequate risk management strategies are essential to understand the complexities of integrating renewable energy sources and mitigate financial losses for investors, thereby ensuring their continued engagement with sustainable energy ventures.

6. Conclusion

This thesis has analyzed the relationship between weather variables, fossil fuel prices, and electricity prices within the Swedish electricity market between 2013 and 2023. The findings demonstrate that wind speeds have a negative correlation with electricity prices, particularly in electricity price area SE4. Natural gas and coal prices were found to be positively correlated with electricity prices. The explanatory power of the independent variables was identified to be higher for natural gas and coal prices in SE4 compared to weather variables, but this influence diminishes for the northern electricity price areas.

7. Limitations

This study is subject to certain limitations regarding the appropriateness of the statistical model employed to draw conclusions on the research question outlined in the thesis. While the model assumes that standard assumptions hold for the analysis, this may not always be the case, potentially leading to violations of the underlying assumptions. In such instances, the validity of the model to draw accurate conclusions is compromised.

Furthermore, the study faces limitations concerning the accuracy of the measured weather data utilized in the report. Given that wind power regions are situated in areas characterized by stable and high wind speeds, there is uncertainty regarding whether the measured wind speeds accurately represent these power-producing regions in Sweden. Consequently, this limitation restricts the conclusions that can be confidently drawn regarding the relationship between wind speeds and electricity.

8. Future research

The potential for further research is vast. Building upon the significant influence of weather variables on electricity prices as uncovered in this study, future research could delve into the potential and efficacy of weather derivatives for mitigating financial risk in Sweden. With the expected growth of renewable energy, poised to play a pivotal role in Sweden's transition towards less carbon-intensive and sustainable energy systems, investigating how these financial instruments can be leveraged to hedge against price volatility arising from weather fluctuations becomes imperative.

Expanding the scope to include additional geographical regions would provide valuable comparative insights. Moreover, the availability of extensive datasets allows for the incorporation of more data, facilitating the utilization of advanced statistical models. For instance, incorporating data from a broader range of weather stations than those utilized in this thesis could enhance the analysis. Additionally, the inclusion of supplementary variables could further elucidate the relationships and causalities between these factors and their impact on electricity prices.

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